Additional file 3: Table S1

Blood metal and nutritional biomarkers did not improve identification of low cognitive performance beyond demographics in older US adults: a pre-registered analysis of NHANES 2011-2014

Shreyan Kancharla, Neil Patel

TRIPOD+AI item 16. The cross-validated performance estimates reported in the main text come from the nested-CV procedure, in which each of the 50 folds fits its own pipeline; no single coefficient set exists from that procedure. This table is a separate refit of the identical pipeline on the complete analytic sample ($n = 1{,}784$, 431 events), reported for TRIPOD compliance and not as a new performance estimate.

C selected by 10-fold CV on the full sample: **0.1** (matches the mode across the original 50 CV folds, $C = 0.1$ in 49 of 50). Intercept: -1.1747. Bootstrap: 2,000 resamples, 2,000 valid (re-selecting C in each resample; a resample is excluded only if it produces a single-class sample or a different one-hot category set than the full-sample fit).

Numeric features are standardised, so each row is the odds ratio per one-SD increase, not per raw unit. **RIDRETH3 and DMDEDUC2 are one-hot encoded without a dropped reference category** (matching the modelling pipeline exactly, which relies on the L2 penalty rather than a reference contrast), so these rows are not standard reference-category odds ratios and should not be read as such.

Table S1: Full M3 logistic regression: standardised coefficients, odds ratios, and 2,000-resample bootstrap 95% confidence intervals.

Feature	Coefficient (standardised)	Odds ratio	95% CI
Age, years	0.572	1.7719	[1.5611, 2.216]
Sex (coded 1=male, 2=female)	−0.18	0.8353	[0.7145, 0.9668]
Income-to-poverty ratio	−0.3108	0.7328	[0.6328, 0.8908]
Blood lead (log)	0.0169	1.017	[0.8803, 1.1725]
Blood cadmium (log)	0.0555	1.0571	[0.9194, 1.2129]
Blood total mercury (log)	−0.039	0.9617	[0.8316, 1.1083]
Blood selenium (log)	−0.0715	0.931	[0.8123, 1.0752]
Blood manganese (log)	−0.0439	0.9571	[0.8412, 1.1021]
Serum vitamin B12 (log)	0.162	1.1758	[1.0302, 1.3963]
Serum methylmalonic acid (log)	0.0906	1.0948	[0.9546, 1.269]
Serum total folate	0.1277	1.1362	[0.9857, 1.4515]
RBC folate	−0.1017	0.9033	[0.7279, 1.0555]
25-hydroxyvitamin D	−0.1193	0.8875	[0.7647, 1.0271]
HbA1c	0.0745	1.0774	[0.9449, 1.2233]
HDL cholesterol	0.0155	1.0157	[0.8527, 1.2226]
Total cholesterol	−0.0745	0.9282	[0.8022, 1.0652]
Triglycerides	0.0501	1.0514	[0.8963, 1.2478]
Serum albumin	−0.0412	0.9597	[0.8359, 1.1036]
Serum iron	−0.0939	0.9103	[0.7976, 1.026]
eGFR	−0.081	0.9222	[0.8026, 1.0756]
Neutrophil-to-lymphocyte ratio	0.1324	1.1416	[0.9793, 1.4352]
Platelet-to-lymphocyte ratio	−0.0653	0.9368	[0.773, 1.1159]
Systemic immune-inflammation index (log)	0.0097	1.0098	[0.8069, 1.2559]
RIDRETH3 = Mexican American	0.0469	1.048	[0.6494, 1.4956]
RIDRETH3 = Other Hispanic	0.3854	1.4701	[1.1422, 2.2946]
RIDRETH3 = Non-Hispanic White	−0.3595	0.6981	[0.5232, 0.919]
RIDRETH3 = Non-Hispanic Black	0.1701	1.1855	[0.919, 1.8108]
RIDRETH3 = Non-Hispanic Asian	−0.0677	0.9346	[0.5933, 1.459]

Table S1 continued

Feature	Coefficient (standardised)	Odds ratio	95% CI
RIDRETH3 = Other/multiracial	−0.1833	0.8325	[0.1675, 1.3136]
DMDEDUC2 = Less than 9th grade	1.0102	2.7462	[2.2624, 4.6369]
DMDEDUC2 = 9-11th grade	0.2881	1.3339	[1.0118, 1.8329]
DMDEDUC2 = High school grad/GED	−0.0889	0.9149	[0.6682, 1.1064]
DMDEDUC2 = Some college/AA degree	−0.5367	0.5847	[0.3774, 0.6851]
DMDEDUC2 = College graduate or above	−0.6808	0.5062	[0.2653, 0.6199]